Om Gururaj Kanade

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.70
Sindhi PU College, Bangalore	Class XII (PUC)	2021	98.83%
Euroschool North Campus, Bangalore	Class X (CBSE)	2019	96.33%

EXPERIENCE

- Asymmetric TF-QKD: Mathematical Modeling & Simulation

Associate Scientist- Intern, Supervisor: Dr. Indrajith V S

- Conducted **mathematical modeling and numerical analysis** for **quantum communication** protocols, translating theoretical formulations into computationally testable models.

- Built **Python/NumPy/SciPy simulation workflows** for parameter sweeps, performance evaluation, and numerical validation of theoretical results.

- Quantum Information, Computing, & Control Theory (QuICCT) Group

Student Researcher, Principal Investigator: Dr. Sangkha Borah

- Researching **quantum machine learning** and computational methods for **non-Markovian AI agents** in **quantum reinforcement learning**.

- **AI/ML Engineering Intern**

Supervisor: Waeez Ahmed

- Applied **AI4Bharat's IndicTrans** model (developed in collaboration with **IIT Madras**) to evaluate and **enhance regional language translation accuracy**, improving performance across **multiple Indic languages.**

TECHNICAL SKILLS

- Programming Languages: C/C++, Python

- Quantum Tools: IBM Qiskit, Google Cirq, Amazon Braket, Pennylane, JAX

- AI/ML & Computing: PyTorch, Scikit-Learn, Hugging Face, Transformers, LLMs, Reinforcement Learning, Git

RELEVANT COURSES

- **AI/ML**: Machine Learning, Deep Learning, Natural Language Processing, Generative AI, Data Strucutres and Algorithms
- **Quantum Technology**: Quantum Computing, Quantum Information Theory, Quantum Communication, Quantum Sensing, Advanced Numerical Techniques for Quantum Algorithms, Mathematics and Computing

PROJECTS

- Non-Markovian AI Agents for Quantum RL

Course: Quantum Computing, Faculty: Dr. Sangkha Borah

- Developed quantum-native RL framework **embedding environmental memory via entanglement**, enabling **history-dependent non-Markovian decision-making** with reward-based optimisation.

- Designed parameterized quantum channels/circuits for agent-environment interactions; applied to **quantum noise**

identification, achieving up to **1.5× accuracy improvement** over conventional Markovian approaches.

- Asymmetric Twin-Field QKD with Actively Odd Parity Pairing

Research Project, Qdit Labs

- Developed an **end-to-end** asymmetric **SNS-AOPP twin-field QKD** framework incorporating channel asymmetry, decoy-state analysis, Chernoff finite-size bounds, and intensity optimization for secure key-rate evaluation.

- Implemented **Python/NumPy/SciPy simulators** for real-time sweeps; derived **AOPP survival rate** as a ratio of mutual information to raw error rate and analyzed the key-rate landscape, revealing **non-monotonic behavior** under fixed intensity settings and the need for **adaptive intensity optimization**.

- Quantum Monte Carlo Estimation with Amplitude Amplification

Course: Computational Lab, Faculty: Dr. Sangkha Borah

- Reformulated classical Monte Carlo estimator into quantum framework using **QAE in Qiskit**, achieving **quadratic error reduction** and **near-constant time complexity** for exponential data growth.

- Benchmarked against classical baselines, showing over **3× improvement** in efficiency and scalability for **probabilistic estimation**.

EXTRACURRICULAR ACTIVITIES

- **Qiskit Advocate**, IBM Quantum, 2026

- Organizing Committee Member, IBM Qiskit Fall Fest 2026 - IIT Hyderabad

- **Member, |QET>,** Quantum Computing Club- IIT Hyderabad, 2026

CERTIFICATIONS AND CREDENTIALS

- Quantum Computing - Centre for Outreach and Digital Education (CODE), IIT Madras, 2023

- Quantum Platform Credentials: AWS Braket · Microsoft Azure Quantum · IBM Quantum